took place. Inflationary expansion amplifies a tiny speck of order in the early universe—it “wound up” the universe to a huge expanse with minimal gravitational entropy—so the 14 billion years of subsequent unwinding, of subsequent clumping into galaxies, stars, and planets, presents no puzzle.

In fact, this approach even tells us a bit more. Just as it’s possible to hit the jackpot on a number of slot machines on the floor of the Bellagio, in the primordial state of high entropy and overall chaos there was no reason why the conditions necessary for inflationary expansion would arise only in a single spatial nugget. Instead, as Andrei Linde has proposed, there could have been many nuggets scattered here and there that underwent space-smoothing inflationary expansion. If that were so, our universe would be but one among many that sprouted—and perhaps continue to sprout—when chance fluctuations made the conditions right for an inflationary burst, as illustrated in Figure 11.2. As these other universes would likely be forever separate from ours, it’s hard to imagine how we would ever establish whether this “multiverse” picture is true. However, as a conceptual framework, it’s both rich and tantalizing. Among other things, it suggests a possible shift in how we think about cosmology: In Chapter 10, I described inflation as a “front end” to the standard big bang theory, in which the bang is identified with a fleeting burst of rapid expansion. But if we think of the inflationary sprouting of each new universe in Figure 11.2 as its own bang, then inflation itself is best viewed as the overarching cosmological framework within which big bang–like evolutions happen, bubble by bubble. Thus, rather than inflation’s being incorporated into the standard big bang theory, in this approach the standard big bang would be incorporated into inflation.

